INCH-POUND

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19 December 1990
SUPERSEDING
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28 December 1984

MILITARY SPECIFICATION

CLOTH, BALLISTIC, NYLON, LIGHTWEIGHT, WATER-REPELLENT TREATED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers water-repellent lightweight nylon ballistic cloth having a specified minimum ballistic resistance.

1.2 Classification. The cloth shall be of the following classes as specified (see 6.2).

Class 1 - Natural
Class 2 - Camouflage Green 483
Class 3 - Woodland camouflage pattern
Class 4 - (Deleted see 6.13)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-T-285	-	Thread, Polyester
V-T-295	-	Thread, Nylon
PPP-P-1133	-	Packaging of Synthetic Fiber Fabrics

MILITARY

MIL-P-46593	-	Projectile, Calibers .22, .30, 50 and 20 MM Fragment-Simulating

STANDARDS

FEDERAL

FED-STD-4	-	Glossary of Fabric Imperfections
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-662	-	Ballistic Acceptance Test Method for Personal Armor Material

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

U S ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-1-1516 - Woodland Pattern - 48 inches
2-1-1516B - Woodland Pattern - 60 inches

(Copies of drawings are available from the U S Army Natick Research, Development, and Engineering Center, ATTN. STRNC-UX, Natick, MA 01760-5017)

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

Method 119 - Color Change Due to Flat Abrasion (Frosting): Screen Wire Method

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard is referenced (see 6.4).

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Nylon. The nylon shall be bright, extra high tenacity, continuous filament (see 6.9).

3.3.2 Yarn. The warp and filling yarns shall be nominal 840 denier, multifilament nylon.

3.4 Color.

3.4.1 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match under incandescent lamplight at 2300 ± 200K.

3.4.2 Color, class 1. The color of the class 1 finished cloth shall be natural.

3.4.3 Color, class 2. The color of the class 2 finished cloth shall be Camouflage Green 483 unless otherwise specified (see 6.2). The color of the finished cloth shall match the standard sample and shall be obtained by piece dyeing using neutral premetallized dyes or neutral dyeing acid dyes, applied at a pH value consistent with the maximum temperature that can be maintained in the equipment used (see 6.7).

3.4.3.1 Colorfastness, class 2. The class 2 finished cloth shall show fastness to laundering (after 3 cycles), perspiration, and light (after 40 hours), equal to or better than the standard sample or equal to or better than a rating of "good". The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not lower than 3.5. Testing shall be as specified in 4.4.3.

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3 4.4 Color, class 3. The color of the class 3 finished cloth shall be the Woodland camouflage pattern obtained by roller or screen printing using four rollers or screens as appropriate for the Light Green 354, Dark Green 355, Brown 356, and Black 357 areas of the pattern (see 6 8 2) The dyeing of the ground shade approximating Light Green 354 shall be accomplished by using a combination of acid dyes and the printing shall be accomplished by using a combination of acid dyes as applicable (see 6 8.1).

3.4.4 1 Colorfastness, class 3 The class 3 finished cloth shall show fastness to laundering (after 3 cycles), perspiration, and light (after 40 hours), equal to or better than the standard sample or equal to or better than a rating of "good" for each of the pattern areas. The finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not lower than 3.5 for all the pattern areas except for Black 357 which shall have an AATCC Chromatic Transference Scale rating of not lower than 1.5. When carbon black is used alone or in combination with other dyes for black 357, the resistance to frosting of the printed cloth shall be equal to or better than the standard sample. Testing shall be as specified in 4.4.3.

3.4.4.2 Pattern execution, class 3. The pattern shall reproduce the standard sample in respect to design, colors, and registration of the respective areas. The pattern repeat of the dyed, printed, and finished cloth shall be 27.25 + 1.25 -2.50 inches in the warp direction. Each pattern area shall show solid coverage; skitteriness exceeding that shown by the standard sample in any of the printed areas will not be acceptable. When the standard sample is not referenced for pattern execution or design, a pattern drawing shall be provided, and the pattern on the finished cloth shall match that of the drawing (see 2.1, 6.2, and 6.4).

3.4.4.3 Spectral reflectance, class 3. The spectral reflectance of each color for the Woodland camouflage printed finished cloth shall conform to the requirements specified in table I, when tested as specified in 4.4.3.

TABLE I. Spectral reflectance requirements, class 3

Wavelength, Nanometers (nm)	Reflectance values (percent)					
	Black 357		Light Green 354		Dark Green 355 and Brown 356	
	Min.	Max.	Min.	Max.	Min.	Max.
600	-	10	8	20	3	13
620	-	10	8	20	3	13
640	-	10	8	20	3	13
660	-	10	8	22	3	13

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TABLE I Spectral reflectance requirements - Class 3 (cont'd)

Wavelength, Nanometers (nm)	Reflectance values (percent) Black 357 Min.	Max	Light Green 354 Min	Max	Dark Green 355 and Brown 356 Min.	Max
680	-	10	8	36	3	22
700	-	10	14	60	8	46
720	-	10	26	78	20	66
740	-	10	40	90	30	80
760	-	10	50	92	32	88
780	-	10	55	92	32	90
800	-	10	55	92	32	90
820	-	10	55	92	32	90
840	-	10	55	92	32	90
860	-	10	55	92	32	90

3.5 Physical requirements. The finished cloth shall conform to the physical requirements specified in table II when tested as specified in 4.4.3.

TABLE II. Physical requirements

Weight, ounces per sq yd (min)	(max)	Yarns per inch (minimum) Warp	Filling
7.7	8.3	33	32

3.5.1 Weave. The weave shall be plain.

3.5.2 Width. The width of the finished cloth shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of the woven selvages when fly-shuttle looms are used and exclusive of the lock in selvages when shuttleless looms are used The edges shall be firmly secured to the body of the cloth and shall not ravel when outward pull is exerted

3.6 Finish. The cloth shall be thoroughly scoured (see 6.5), heat-treated (see 6.6), and water-repellent treated, and shall be processed so as to meet the ballistic resistance requirements of this specification (see 3.10).

3.7 Water-repellency. The cloth shall be given an approved water-repellent treatment (see 6.10).

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3.7.1 Dynamic absorption. The dynamic absorption of the finished cloth shall not exceed 20 percent when tested as specified in 4.4.3.

3.7.2 Resistance to organic liquid. The finished cloth shall not show wetting by n-dodecane when tested as specified in 4.4.3.

3.8 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.3.

3.9 Shrinkage. The finished cloth shall not shrink more than 3.0 percent in the warp and not more than 2.0 percent in the filling when tested as specified in 4.4.3.

3.10 Ballistic resistance. The ballistic limit V_{50} for 16 layers of unbonded cloth shall be 1225 feet per second when using a 10 shot V_{50} with a maximum spread of 125 feet per second. Testing shall be as specified in 4.4.3.

3.11 Length and put-up. For Government procurements only, unless otherwise specified (see 6.2), the cloth shall be put-up in full width rolls as specified in 5.1. Each roll shall be not less than 80 yards in length and shall contain not more than two pieces. No single piece shall be less than 40 yards in length.

3.12 Face identification. The face side of the cloth shall be identified by applying a stamping on that side of the cloth with the word "Face" at each end of the roll.

3.13 Fiber identification. Each roll of the finished cloth shall be labeled or ticketed for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.14 Workmanship. The finished cloth shall conform to the quality of product established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the applicable established maximum point values.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1 1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 thru 4.4.2.4 and tested for the characteristics specified in table III.

4.4 Quality conformance inspection.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the cloth conforms to the requirement specified in 3.3.1.

4 4 2 End item visual examination

4 4.2.1 Yard-by-yard examination. Each roll in the sample shall be examined on the face side only. When the total yardage in the roll does not exceed 100 yards, the entire yardage in the roll shall be examined. When the total yardage in the roll exceeds 100 yards, only 100 yards shall be examined. All defects as defined in section III of FED-STD-4, which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.4.2.1.1. No linear yard (increments of 1 yard on the measuring device of the inspection machine) from any one roll within the sample shall be penalized more than four points. The sample size shall be in accordance with the following:

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Lot size (yards)	Sample size (rolls) 1/
3200 or less	8
3201 up to and including 10,000	13
10,001 and over	20

1/ No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls in which case all shipping containers shall be present in the sample. The lot shall be unacceptable if the points per 100 square yards of the total yardage examined exceeds 30 0 points. The lot shall be unacceptable if the points per 100 square yards of two or more individual rolls exceed 45.0 points. If one roll exceeds 45.0 points per 100 square yards, a second sample of the size indicated above shall be examined for individual roll quality only. The lot shall be unacceptable if one or more rolls in the second sample exceeds the 45 0 points per 100 square yards. Point computation for lot quality and individual roll quality shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Contracted width of cloth (inches)} \times \text{Total yards inspected}} = \text{Points per 100 square yards}$$

4.4.2.1.1 Demerit points. Dermerit points shall be assigned as follows:

For defects up to and including 3 inches in any dimension	- one point
For defects exceeding 3 inches but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

The following defects, when present, shall be scored four points for each yard in which they occur.

Baggy, ridgy or wavy cloth
Mottled or cloudy (class 2)
Width less than specified
Edge ravels when pulled outward
Slack or tight selvages 1/
Overall uncleanness
Skitteriness (mottled, uneven color) of pattern exceeds that shown by standard sample (class 3)
Pattern design not equal to standard sample (class 3)
Excessive feathering or spew (fuzziness at color boundaries) of pattern as compared to the standard sample (class 3)
Incorrect color in any part of the pattern (class 3)
Pattern repeat not equal to the standard sample (class 3)

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Pattern repeat less than 24 75 inches or more than 28 50 inches (class 3)
Excessive grinning (off register, gap where ground shade shows through) of pattern as compared to the standard sample (class 3)
Excessive hallowing or trapping (overlapping of colors) of pattern as compared to the standard sample (class 3)

1/ To determine the presence of unacceptable selvage conditions, the following procedure shall be observed: During the visual examination, the perch shall be stopped a minimum of three times for each roll in the sample, the tension removed and the ballistic cloth examined for the selvage conditions. Suspect rolls should be removed from the perch and unrolled on the floor or a table and further examined for the condition. A waviness in the selvage or significant ripples diagonally across the width of the fabric is an indication of slack or tight selvages.

4.4.2.2 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for the defects listed below. If two or more rolls contain a length defect, the lot shall be rejected. In addition, the lot shall be rejected if the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the roll ticket.

Length defects

Length of roll less than minimum specified
Length of roll is more than 2 yards less than the length marked on the ticket
More than two pieces on a roll
Length of individual piece is less than 40 yards

4.4.2.3 Shade and appearance examination. During the yard-by-yard examination of the dyed or dyed and printed cloth, each roll in the sample shall be examined for shade and appearance. Two or more rolls in the sample off shade or shaded side to side, side to center, or end to end, or any roll that does not have the same appearance as the standard sample shall be cause for rejection of the entire lot represented by the sample.

4 4.2.4 Roll identification examination During the yard-by-yard examination each roll in the sample shall be examined for the defects listed below. The lot shall be unacceptable if two or more rolls in the sample contain one or more of the following defects:

Face identification missing from either or both ends.
Not labeled or ticketed in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

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4.4.3 End item testing. The cloth shall be tested for the applicable characteristics indicated in table III The methods of testing specified in FED-STD-191 wherever applicable and as specified in table III shall be followed Except for ballistic resistance, the physical and chemical values specified in section 3 apply to the results of the determinations made on a sample unit for test purposes as specified in the applicable test method. For the ballistic resistance characteristics, the lot average of all sample units in the sample shall apply. Except for ballistic resistance, the lot shall be unacceptable if one or more sample units fail to meet any requirement specified. For ballistic resistance, the lot shall be unacceptable if the lot average value fails to meet the minimum ballistic resistance requirements (see 3.10) The sample unit shall be as follows:

a. For ballistic test. Six full width cuts, each 15 inches long of the finished cloth and originating from a different roll.

b. For all other tests. One cut 3 yards long full width of the finished cloth originating from one of the six rolls from which the sample unit of the ballistic test is drawn.

Each individual cut used in the sample unit, shall be marked to indicate contractor's piece, lot, and roll number, and the Government lot number. The sample size for both ballistic and other tests shall be in accordance with the following:

Lot size (yards)	Sample size (No. of sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE III. End item testing

Characteristic	Requirement paragraph	Test Method
Dye identification (class 2)	3 4.3	1/
Colorfastness to (classes 2 and 3):		
Laundering (after 3 cycles)	3.4.3.1 and 3.4.4.1	5614 2/ 3/
Perspiration		5680
Light (after 40 standard fading hours)		5660
Crocking		5651

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TABLE III End item testing (cont'd)

Characteristic	Requirement paragraph	Test Method
Resistance to frosting (class 3, Black 357 only 4/)	3.4 4 1	AATCC method 119 5/
Spectral reflectance (class 3)	3 4 6	4.5.4
Weight	3 5	5041
Yarns per inch	3.5	5050
Weave	3.5 1	Visual 6/
Finish: Scouring and heat treatment	3.6	1/
Water repellency	3.7	7/
Dynamic absorption	3.7.1	5500
Resistance to organic liquid	3.7.2	4.5.1
pH	3.8	2811
Shrinkage	3.9	4.5.3
Ballistic resistance	3.10	4 5.2

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement

2/ The specimens must be dried after each of the three laundering cycles

3/ Only the stain on the nylon and cotton fiber of the color transfer cloth shall be evaluated.

4/ When carbon black alone or in combination with dyes is used for Black 357.

5/ Except that the number of abrasion cycles shall be 900.

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6/ One determination per sample unit and the result reported as "pass" or "fail"

7/ The contractor shall report the water repellent formulation used, and certify that no other material has been added.

4.4.4 Packaging inspection. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with requirements of PPP-P-1133.

4.5 Methods of inspection.

4.5.1. Resistance to organic liquid test. Place a small specimen of the cloth on a smooth horizontal surface face up. Using a pipette, or eye-dropper, gently deposit one drop of n-dodecane on the surface of the specimen. After 1 minute, examine the specimen under light at an angle. Absence of light reflectance at the fabric-drop interface shall be taken as evidence of wetting. Three specimens (or areas) taken at various locations across the sample shall be tested. Evidence of wetting on one or more specimens shall be considered cause for rejection of the lot represented by the sample.

4.5.2 Ballistic resistance test. The test panels shall be made up from the six cuts (see 4.4.3), five shall be cut into three parts and a 15-inch by 15-inch specimen shall be taken from the sixth cut to make up sixteen layers required for a test panel. The size of each layer in the panel shall be approximately 15 inches by 15 inches. The specimens shall be tested in accordance with the procedure in 4.5.2.2. Prior to conducting the test, the panels shall be preconditioned in the ballistic test area for at least 24 hours with air freely circulating on all sides of the layers. All test panels shall be sewn in accordance with 4.5.2.1.

4.5.2.1 Sewing of ballistic test panels. All ballistic test panels shall be sewn around the periphery 1/2 ± 1/8 inch from the edges. The sewing shall be accomplished in accordance with the following:

a. Thread-Nylon, conforming to type I, class B of V-T-295 or polyester conforming to type I, class 1, subclass B of V-T-285. Thread size shall be "E". Color shall be optional.

b. Stitches per inch - 8 to 10.

c. Seam type - SSa-1 of FED-STD-751.

d. Stitch type - 301 of FED-STD-751.

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4 5 2 2 Ballistic resistance V_{50} procedure. The ballistic resistance test shall be conducted in accordance with MIL-STD-662 with the following exceptions:

a The projectile. The test projectile shall be the 22 caliber type 2 projectile for body armor as specified in MIL-P-46593.

b Barrel. The barrel shall be a .30 caliber rifled barrel with a one-in-sixteen twist. The barrel length shall be 28 inches. The barrel shall be chambered to accommodate firing the specified sabot.

c Sabot. A sabot will be used in loading and firing the projectile The sabot shall be manufactured to conform to figure 1. The projectile is to be loaded in such a manner that the whole body of the projectile is on the sabot except for the gash ring The projectile shall be loaded so that the base of the projectile is in contact with the pusher plate. The impact shall be considered fair when an unyawed test projectile strikes an unsupported area of the test panel at least 2 inches from a supported area of the test panel and at least 2 inches from any previous impact, and shall not involve the same warp of filling yarns of any previous impact. The V_{50} limit for the lot shall be reported as the average of all panels tested from the lot.

d. Pusher plate. A pusher plate shall be utilized and placed flush with the edge of the throat of the cartridge. The pusher plate shall have the following characteristics:

1. Diameter - 0.304 inch (reference dimension)
2. Thickness - 0.050 inch (reference dimension)
3. Material - Aluminum, 7075T6 or 6061T6

e. Yaw card measurement system. A yaw card shall be used to determine yaw. The following procedure shall be employed.

1. Place the yaw card directly in front of the test sample with the emulsion side facing the sample. (The yaw card will be devoid of any markings. Kodak photographic paper, single weight, Kodabromide, or equal may be used for the yaw card.)
2. After the test, carefully measure the yaw card to determine the largest dimension of the hole caused by penetration of the projectile. An optical magnification device with a magnification between 5X and 10X shall be used for making this measurement. Compare this measurement to the X-axis on figure 2. The Y-axis of figure 2 will be used for determining the degree of the yaw.
3. The firing is considered invalid if the determined degree of yaw is more than 5°.

f Test sample mount. The armor test sample shall be secured with impact sites perpendicular to the line-of-flight of the projectile. The frame supports and clamps or mounting fixtures must be capable of retaining the

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sample and withstanding shock resulting from ballistic impact by the test projectiles. The test sample mount shall be capable of adjustment for moving the sample in the vertical or horizontal directions so that the point of impact can be located anywhere on the sample, and rotation on the vertical axis is so that zero degree obliquity impacts can be achieved anywhere on the sample. The test sample mount shall be able to rotate on the vertical or horizontal axis so that various obliquity attack angles can be achieved. The test samples shall be mounted with a rigidity equal to or greater than the actual installation of the part. The test panel shall be mounted between an inner and outer metal frame. The frame shall be designed to capture the entire periphery of the test panel to a depth of 1-1/2 to 2 inches. The frame shall also be designed to provide a minimum available test area of 121 square inches. The mounting frame shall be clamped, at a minimum, in eight locations around its periphery, i.e., at each corner and at the mid-point of each side. (NOTE: vise-grip type pliers or equal have been found to be acceptable for clamping.) The test panel shall be mounted with the expressed intent of providing uniform tension throughout and to provide for a minimum of deflection at projectile impact.

g. Measurement of yaw. Yaw shall be determined using the system specified in "e". Yaw may be examined at any time deemed necessary. However, in any event, yaw shall be measured at intervals no less than those indicated below. When the barrel is new, a minimum of 25 shots shall be fired to "break-in" the barrel. Measurement of yaw shall be five successive shots at each interval.

Measurement interval

1. After "break-in" period.
2. After 250 shots.
3. After 500 shots.
4. After 750 shots.
5. After 1000 shots.

In the event the shots fired indicate yaw, the barrel should be rechecked for effect on yaw as follows: Fire five shots, if three or more shots exhibit yaw, discard the barrel. If one or two shots exhibit yaw, investigate the cause and correct. If no yaw is indicated at the 1,000-shot interval, measurement for yaw shall be made at 100-shot intervals thereafter until such time as yaw is indicated. In the case of a dispute concerning a particular barrel, yaw shall be measured by a photographic measurement system using a multiflash light source to determine projectile velocity and yaw. Yaw shall be measured by the system to an accuracy of 0.5 degree.

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h. Fair impact. The impact shall be considered fair when an unyawed test projectile strikes an unsupported area of the test panel at least 2 inches from a supported area of the test panel and at least 2 inches from any previous impact, and shall not involve the same warp or filling yarns of any previous impact.

4 5 3 Shrinkage test. Three specimens, each 20 inches by 20 inches, shall be cut from the cloth and conditioned for 4 hours at standard conditions. Each specimen shall be marked with an 18-inch square and formed into a loop by stapling two opposite sides together. A glass rod 1/2 inch in diameter and 21 inches long shall be placed inside the loop at the top and a glass rod 1/4 inch in diameter and 21 inches long, weighing approximately 100 grams shall be placed inside the loop at the bottom. Each specimen shall be immersed in a tub of boiling water, by attaching twine or wire to the ends of the top glass rod, and allowing the specimen to hang freely in the boiling water for 15 minutes. The specimens shall be removed and allowed to drain. After the staples are removed, the specimens shall be dried in a flat position on a horizontal screen, and reconditioned as stated above. The marked square shall be measured in three places in the warp direction and three places in the filling direction and the percent of shrinkage determined The shrinkage of the sample unit shall be calculated as follows and reported to nearest 0.1 percent.

$$\text{Shrinkage, percent} = \frac{A - B}{A} \times 100$$

A = Average of initial measurements (3 specimens).

B = Average of measurements after laundering (3 specimens).

The three specimens shall be selected from the cloth sample unit as follows: One specimen from each side of the cloth to within 3 inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns. Each elongation value shall be prefixed with a minus sign and both enclosed in parenthesis.

4.5.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers (nm), at 20 nm intervals on a spectrophotometer (see 6.11) relative to a barium sulfate standard, the preferred white reference standard. Other white reference materials may be used, provided they are calibrated to absolute white; e.g., Halon, magnesium oxide, or vitrolite tiles (see 6.12). The spectral band width at 860 nm shall be less than 26 nm. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either CIE Source A or CIE Source D65. The specimen shall be measured as a

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single layer, backed with six layers of the shade cut from the standard. Readings shall be taken on a minimum of three different areas and the data averaged. The specimen shall be viewed at an angle no greater than 10° from normal. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nm. When the measured reflectance values for any color at four or more of the listed wavelengths do not meet the limits specified in table II, it shall be a test failure.

5 PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Levels A and Commercial preservation. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Levels A, B and Commercial packing. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with PPP-P-1133 as applicable.

5.4 Special marking. Each roll shall have the type of nylon used, marked on the ticket, i.e., Nylon 6,6 or Nylon 6.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nylon ballistic cloth is intended for use in the manufacture of body armor, helmets, and armored clothing.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. When color other than Camouflage Green 483 is required for class 2 (see 3.4.3).

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f. Pattern drawing if required (see 3.4.4.2).
g. Width of cloth required (see 3.5.2).
h. Length required, if other than specified (see 3.11).
i. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Scouring. Thorough scouring of the cloth will be found to be necessary. Where caustic soda or soda ash have been used in the scouring process, good results have been reported. The use of surface active agents or soaps is not recommended. Solvents and water extractable matter should be held to a minimum.

6.6 Heat treatment. Excessive time or temperature of heat treatment or stretching or stressing of yarn or cloth may reduce the ballistic resistance of the cloth.

6.7 Class 2 dye combination. A suggested but not mandatory dye combination for Camouflage Green 483 is as follows:

Camouflage Green 483

Acid Blue 171
Acid Orange 162

6.7.1 pH and temperature. The standard sample for class 2 was dyed at a pH of 6.8 in a closed jig. The contractor is cautioned that the relationship between temperature of dyeing and pH of the dye bath is critical and may affect the ballistic performance of the fabric. Research has indicated that the higher the temperature at which dyeing is accomplished, the higher should be the pH and that lower values of pH are reasonably safe as the temperature of dyeing is lowered. For example: at 170°F, a pH of 4 can be used; at 190°F, a pH of 5 is adequate; at 200°F, the pH may fall to between 5.5 and 6.0. In a completely closed jig where temperatures of 212° to 215°F are possible, the pH of 6.8 is adequate to achieve exhaustion and maintain ballistics. A pH value below the value or 4.0 should not be used under any temperature condition as this will deteriorate the ballistic performance of the fabric.

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6 8 Class 3 dye combination

6 8 1 Ground shade class 3 The following dyes have been found acceptable for dyeing the ground shade approximating Light Green 354.

Acid Orange 156
Acid Blue 258
Acid Red 266
Acid Yellow 219

6 8.2 Pattern shades, class 3 The printing of the Light Green 354, Dark Green 355, Brown 356, and Black 357 colors may be accomplished by varying the amount of the following dyes:

Acid Orange 156
Acid Red 266
Acid Blue 258

Shade Black 357 may also be satisfactorily printed with the use of carbon black alone or in combination with the above dyes.

6.9 Nylon. Some nylon types that have met the requirements of 3.3.1 are:

Nylon 6	-	Produced by Allied Chemical Co. - Type IT-70
Nylon 6,6	-	Produced by E. I. DuPont De Nemours Co. - Type 728
Nylon 6,6	-	Produced by Monsanto - Type AO7

6 10 Water-repellent treatment. The water-repellent treatment consists of the co-application of an emulsified fluorocarbon and a modified melamine hydrophobe type of repellent. Approval of such components and combinations is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5019 and is based on more extensive tests, including those for toxicity, which are not set forth in this specification. Because of the time necessary to conduct full evaluation (approximately 6 months), only those chemical treatments already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

6.11 Spectrophotometers. Suitable spectrophotometers for measuring spectral reflectance in the visible/near infrared are the Diano Hardy, Diano Match Scan, Hunter D54P-IR and Macbeth 1500 with IR options.

6.12 White standard. Barium sulfate of suitable quality for use as a white reference standard is available from the Eastman Kodak Company. The same source has available, magnesium reagent (ribbon) and Halon. Suitable tiles can be obtained from the National Bureau of Standards or the instrument manufacturers.

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6.13 Supersession data The class 4 woodland camouflage pattern, 100 percent pigment printed cloth has been deleted since it is no longer required.

6.14 Subject term (key word) listing.

Armored clothing
Body armor
Bullet protection
Fragment protection
Helmets

6.15 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

(Project 8305-0326)

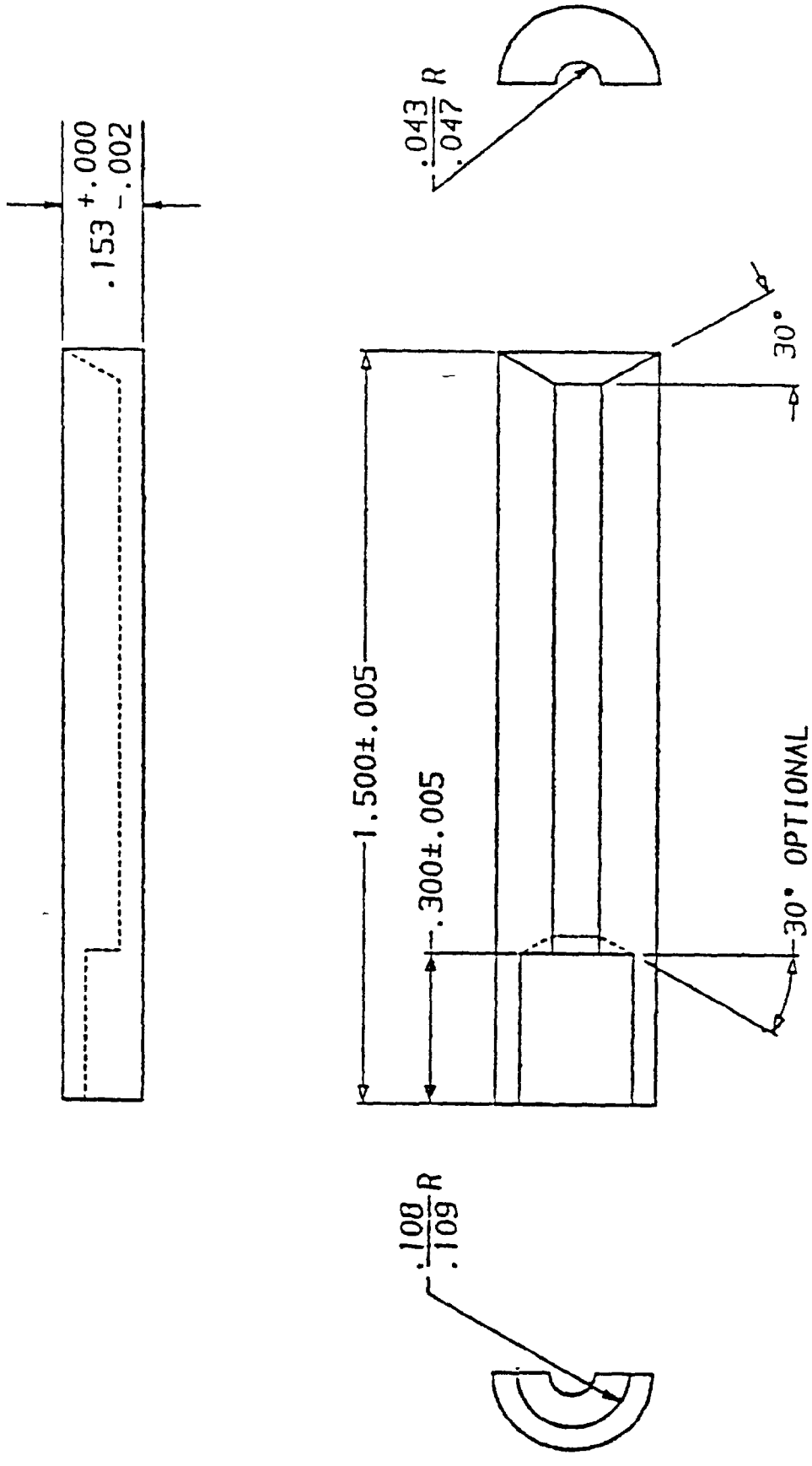
Review activities:

Army - MD
Air Force - 82
DLA - CT

User activity:

Air Force - 45

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8-2-002

FIGURE 1. SABOT FOR 17 GRAIN FSP (FRAGMENT SIMULATED PROJECTILE)
INJECTION MOLDED, TWO HALVES
MATERIAL: POLYCARBONATE
DIMENSIONS ARE IN INCHES

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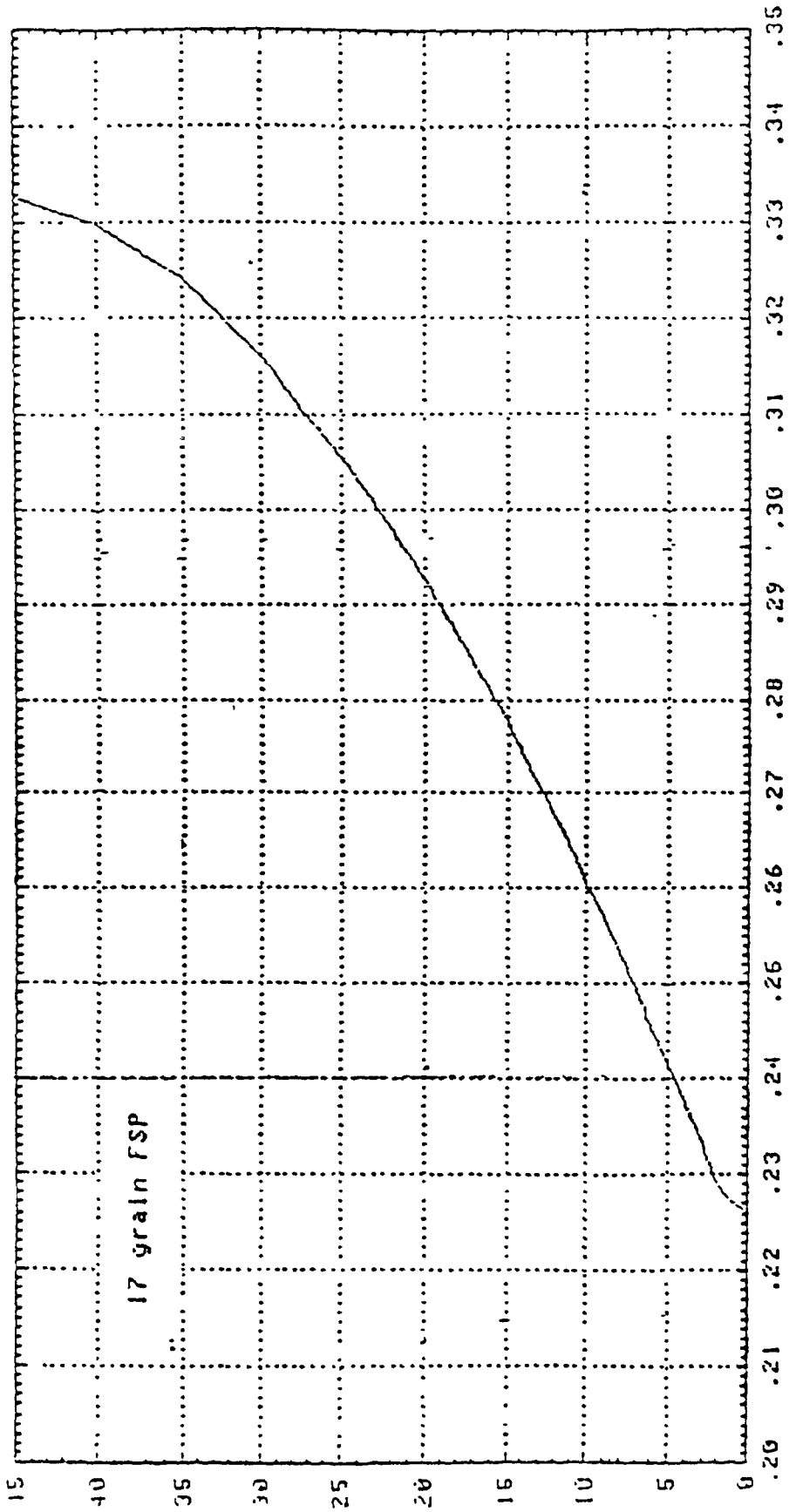


FIGURE 2 PROJECTILE YAW VERSUS MAXIMUM HOLE DIMENSION

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER	2 DOCUMENT DATE (YYMMDD)
	MIL-C-44031B	1998 December 10

3 DOCUMENT TITLE
CLOTH, BALLISTIC, NYLON, LIGHTWEIGHT, WATER-REPELLENT TREATED

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

NAME	b. TELEPHONE (Include Area Code)	
	(1) Commercial	(2) AUTOVON
J.S Army Natick RD&E Center	508-651-4532	256-4532
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	

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